

ActInf Livestream #020.2 ~ The Emperor's New Markov Blankets

Livestream | The Emperor's New Markov Blankets | 1:55:07

hello and welcome

everyone to actin flab this is actin

flab

live stream number 20.2 this is our

second

group discussion on the paper the

emperor's new marco blankets and today

is april 27

2021 welcome to

actimflab everyone we are a

participatory

online lab that is communicating

learning and practicing applied

active inference you can find us at our

links on this

slide this is recorded in an archived

live stream

so please provide us with feedback so

that we can improve on our work

all backgrounds and perspectives are

welcome and we'll be

following video etiquette for group live

streams like

muting if we're not going to be talking

and raising our hands so that we can

hear

from everyone we're down here on the

bottom of the slide

in the second discussion for the

emperor's new markov blankets

and closing out april with this really

great discussion so check out this link

rb.gy/kvnpyc

if you want to see what's upcoming for

the different streams

the goal of today is really just to
continue

this awesome exploration and learning
and discussion that we've been having
and really appreciative that the authors
are

here to be talking about it with us so
we'll be able to

ask some cool questions and anyone who's
watching feel free to ask a question in
the live chat and we'll

relay it to the group today we're just
going to be

going wherever we need to go we'll start
with introductions

and then if people want to call out a
specific figure

or a specific question uh or especially
reference a quotation in the text
or just something that they've prepared
that would be really great

so we'll just go around and introduce
everyone and

say hello feel free to just introduce
yourself however you'd like

and give a short check in and then after
we

have an introduction round we can return
to these warm-up questions

and anyone can feel free to chime in on
one of these questions

such as what they're excited about so
i'm daniel

i'm in california and i will pass it to
shannon

hey everyone i'm shannon i'm usually in
california but i'm in south dakota

for the pandemic and i will pass it

to sweden hello i'm stephen i'm in
toronto

and i'm doing a practice-based phd
through the solomon school
of applied psychology and i will pass it
to scott

hello um scott david from seattle
washington

and at the university of washington
applied physics lab

and uh we're working on synthetic
intelligence and i will pass it to
wait i lost my list

who hasn't chatted yet uh dean have you
had a chance

hi uh i'm dean i'm from calgary
i'm retired and i just

like hanging out here because i learned
a few things

and i'll pass it to chris

hi i'm chris i'm a

postdoc at the rural universitat in
bochum in germany but i'm joining from
berlin today and i'm one of the authors
on the paper

and i'll pass it to joe

uh thanks chris i'm jo dewhurst i'm a
postdoc at the munich

center for mathematical philosophy also
one of the authors on the paper

and my background's in philosophy of
neuroscience and computer science
primarily i'd say

thanks i'll pass together

hi everyone uh happy to be here uh again

i'm yada raiber also one of the authors
of the paper

i am currently based in amsterdam

although i'm supposed to be in
uh sydney australia um
yeah uh happy to be here uh today
today's a birthday at the dutch
king so it's a special day happy
birthday
okay i think that's everyone with the
introduction so anyone
please feel free to raise your hand and
start off some of the warm-up questions
are what is something that you're
excited about today could be related to
the
paper or tangentially what's something
that you liked or remembered about the
paper
or from our discussion last week and
what is something that you're
wondering about or you'd like to have
resolved by the end of today's
discussion
maybe another question would be what
kind of momentum
or what kind of next steps would make
sense you know
emerging from this whatever we end up
resolving or not
in the next conversation where will that
take us
in our research or in our practice
journeys i guess
so joe and then anyone who'd like to
speak
uh thanks um do i need to lower my hand
again or is that done
uh it's just good good great uh cool so
i missed the discussion last week so
some of this may have already been

covered but
one thing which relates a little bit to
something we were discussing just before
we went live here and i think is kind of
um a broadening of the points we wanted
to make in the paper which i which i
wonder about
is uh what what the
if if active inference can be kind of
arbitrarily applied to any given case
given the right kind of setup and
parameters and so on
um what does this mean for its future
application
is this so far as it's too general or is
this a strength insofar as it can be
used
anywhere really um and this this i think
is kind of lurking underneath some of
the points we make in the paper
that's a more general
cool so can active inference apply to
anything and everything would that be a
strength or would that be
a weakness awesome question
any other thoughts anyone wants to
raise your hand yet steven
i'd be interested to um learn more about
the way that
the pearl blankets and the fristen
blankets
um interpretations might be useful
in applications and how that might
inform you know the way that modeling's
done
so that that's kind of exciting for me
so that's one thing that i'm i'm looking
forward to

great question as well the um
distinguishing different kinds of
blankets and even pulling back to like
what kind of metaphors will we use and
how will we relate technical terms
to different settings different
practices those are some
big questions we want to address scott
and then anyone else
yeah i'm interested in all in both of
those questions or all that
aspects and then also the frizzle of the
paradox of those two
questions so it's a chicken and egg
question of first and
enough first of active inference in a
sense
right we're observing something we're
projecting it out as a teleology onto
the world
and then asking if it's a good teleology
right so it's so in a way it's like if
you go for david bowm idea that
all reality is first a thought that
includes the realities that we don't
create entirely but that we project
so to one of the questions i have so as
a i was a practicing lawyer for 30 years
or 25 years ish and when i see active
inference
i think contracts and so i think of it
as biomimicry
of what we can do socially and project
further with real intention to control
systems
so that so what i'm wondering is and i
think both are valid
but because so ultimately is it a notion

i'll use the word biomimicry but
obviously i don't mean in the physical
sense
but from us to learn from deep time
what we can do as biological organisms
to better
organize ourselves despite our social
elements so to get back to our biology
in a way and and
maybe organize ourselves a bit better
and more resiliently
with reference to that anyway it's a
that's the frizzle i'm interested in is
the both the cause
being upward and downward with regard to
active inference
thank you cool and
connecting the micro and the macro it's
sort of
sometimes it's laid out abstractly like
multiple spatial scales that are nested
within each other or time scales
but then when we talk about action and
we have the example of
sort of the person in a maze and you
take a step back so that you can go
around
that's the kind of settings where we
might want to
apply active inference where the initial
action
might be non-linear or away from the
goal
in a sense but it's part of a policy
trajectory that actually gets you where
you want to be so and the other
the other part of it is with the markov
blankets being a bayesian

notion because of the beige then you say
oh when does bayes
when is bayes effective well that
becomes a pretty broad
application right because i always say
to people to get them in the space i
always say if you're in a
dark forest with no flashlight you don't
run full speed ahead
you stick out your hand in front of your
face right and that's basically active
inference
as i say you don't you know you want to
know what's in front of you before you
run into a tree
right so anyway good stuff thank you
any other yep yellow and then anyone
else
yeah so uh i think last week we ended
the
uh discussion on shannon's point on what
it means to have a folk
book psychological form of active
inference and i think it related to the
uh to the tension between needing to
have a very deep mathematical
understanding of an increasingly
kind of developing field of mass and
applying
these kind of more broad set of ideas uh
to a particular kind of case so i
thought this was a a point to explore
further and and think about how this
would look like this uh this
more folk psychological way of
understanding active inference
using the concepts maybe without the
nitty-gritty details of mathematics

shannon yeah

um i was hoping that we could bring that
up today

and maybe ask joe's take um and

also sort of if we're if we're talking

about something like

a fristen blanket and that's supposed to

be a real demarcation of a system

or a pearl blanket which is supposed to

just be

how we describe the system

um then maybe in this you know folks

psychological notion

of active inference or talking about how

a phenomenon appears to

this person in a crowd versus another

person in a crowd where both phenomenons

are real

but depending on which one you decide to

model you'll get a different picture of

how the crowd's dynamics are changing

and we could

model that mathematically but just as

easily someone in the crowd could have

an informed notion of what it is that

they're

receiving and how it seems that the

markov blankets

however they are are emerging around the

crowds in certain ways

and you could start having a discussion

that

maybe you can you know translate from

the scientists doing the fancy math

and the person in the crowd who's just

existing

thanks very interesting steven than

anyone else

yeah just building on that i think i'm
also
curious because there seems to be
there's two sides there seems to be this
primordial soup which is referred to in
the paper
as well and the variational free energy
and how that can
self-organize um and then and that
happens at smaller distances
and then these longer distances where
you've got expected free energy because
the organism is seen
into the environment with sensory
information
and often it's used in these partially
observable observable markov decision
processes
for modeling that and i'll be just
curious if
those two domains of modeling which seem
to come up and i know there's more but
those two seem to come up a lot in the
literature how that pans out maybe with
some of these questions of
you know is there an intuitive feeling
of how
the morphology is moving towards a
target which could be that deeper sort
of knowing of the body
and then there's this kind of partially
observable decision process which is
kind of cognitive
sort of looking in the world and which
ones are more or less
like markovian in some ways and how does
that affect how we try and
uh play with them which is

or model them or play with them which i
think shannon was saying in a way is
in a good way is exploring these ideas
phenomenologically
thank you scott and then anyone else
the stevens notion got me thinking about
the notion of glass
in glass they frequently talk about
first order second order third order
order um and so you have that amorphous
material it's semi-amorphous solid right
so it's not really crystalline
not really liquid so it has these orders
that are discernible through different
framings of
maths and it made me think about stephen
when you were saying that cognitive
glass
in a sense that we have this orders of
order that we project onto
disorder and so we those
teleologies ultimately roll up into
things like sovereignty
which is seen as perfect order in the
abstract right the sovereign does not
need to ask for permission or
forgiveness
because it is the quintessence the fifth
element
right it is not earthly so we have this
metaphysics that happens at the
bottom or top of the turtle stack
depending on which direction you're
going
and so and i think that's kind of nice
here hey blue
i think that's kind of nice here when we
look at order

and abstraction and the last point about
it is that
you know kandinsky the artist said that
violent societies yield abstract art
and so one of the things i always
wondered is is abstraction the reverse
of violence
so if we look at sovereignty as being a
total abstraction
as i just described maybe sovereignty
is also total violence and mills said
that
sovereign has the monopoly of legitimate
violence
so anyway those are few ideas about
maybe we're looking at cognitive glass
here with different
orders of order that we can
project onto the system and maybe that
helps us
when those when we look at that order as
levels of abstraction as you get further
from the first and second order into the
third and fourth order
maybe it starts to be something that
starts to yield a more
projective or teleological element thank
you
thank you scott dean and then anyone
else
yeah so i i wasn't part of the
the tile formation last week i was kind
of in the background and then
lost it but i did go back and watch the
the whole thing and
i'm hoping that uh chris has a few more
things to say today about construal
because um

that to me is kind of the the inflection
point and

he was starting to talk about some of
those things and the separation between
uh what did he have model and target

i think at the end of the day um if we
were to give a lay person or some i
think the

term folk has been used i think if we
give them a mark off blanket and a pearl
blanket

um a lot of people will go how does that
change my basic question which is how do
i get the world

i think that's what all of us here are
trying to do and i think

for a lot of people who aren't even
aware of what a markup like it is
or free energy principle is they still
want that basic question answered
so i'm really interested to find out
today

how the paper moves us to a place where
that becomes easier

that the obstacles to getting the answer
to how do i get the world
suddenly flatten and become

i don't know more familiar

thanks dean and a thought on the
cognitive glass and what you just said
there so framing it in big terms about
sense making sense making about the
world as well as about

active inference is really helpful and
scott when you mentioned the cognitive
glass

i thought about oobleck it might be
called different things different places

but it's cornstarch and water
and so when you slap it really hard it's
like a resilient boundary and your hand
comes away dry and it's like you're
hitting a piece of
concrete but then if you go very slowly
it kind of is like a liquid so it's an
amorphous
solid like the glass now the boundary
between the air
and the oobleck is something that is
real about the world
and so there's a distinction there but
it's almost like depending on how the
boundary is crossed
not even like two different types of
things they're both touch
but it's actually the dynamics of the
stimuli that dictate
whether the response is going to be very
stubborn and more like a solid
or whether it will be very giving and
again it's not like it's light versus
sound
to different sensory inputs this is
different quantitative differences
within one interface modality and it
just shows how even when there's a clear
boundary in the world we
think like between um two phases still
the behavior of the interface can be
very dependent on the type of
interaction
so scott and then anyone else
made me think is it schenerian all the
way down
or do you have myriad schenerian
vectors intersecting and we see that as

complex

stephen then anyone else

i think you could also see this question

around the glass

because glass at a certain way of

freezing or becoming solid it doesn't

crystallize it doesn't break up it

you can still see through it so you've

got this question of it being part of

the medium

through which information can pass

because you can see through it but

there's also that

question around that goldilocks zone

when

if you want something to rearrange and

be the material like happens in the

primordial soup

kristen talks about there's a you know

too lower temperature it sort of

just becomes a solid too higher

temperature it's going to dissipate and

start to

so you've you've got a bit of the

overlap between you know internal

external solid

liquid um map territory it's kind of

interesting

yes different phases of the mind for

sense making too rigid or too fluid so

um joe then scott and then anyone else

who'd like to speak

uh thanks yeah so this this just made me

think that quite quite a few of these

points for me at least come back to a

kind of central

thing so so shannon's question about

folk psychology which i'm very

interested in or kind of folk
inference um questions about teleology
and
uh questions about you know whether
whether it's just something simple all
the way down which makes it look like
it's complex um
to me makes me think there's an
important question here about the role
of
um perspective and kind of viewing of
these systems going on and
and the way i see a lot of a lot of the
more teleological things that we project
onto these systems
um some even even like a boundary
perhaps it isn't really something we
should uh take to be in the system that
is rather just something we're
uh kind of modeling the system as having
so so it comes back to my target model
question as well um
so just just a central question here of
what we take to be really in the world
and what we take to just be
part of our understanding of the system
of a world
cool thank you scott and then anyone
else
i love the conversation because everyone
is moving
so nicely and easily and
with such uh facility back and forth
between the metaphysics and the physics
and appropriately and i don't i don't
mean this is not a diminishment right
because that
we're we're demarking those edges one of

the things here that occurred to me in
the last couple of notes is the notion
of seed crystals
because we are talking about phase
changes in system
behavior i think there's very often a
threshold
kind of trigger type of thing that
happens and so
that notion that stephen was bringing up
about phase
seems interesting and that was implicit
in what you were saying about cornstarch
as well
and so one of the things i think is
interesting is what do those
triggers look like and how are those
encoded
and in so when you're talking about
popular encoding
versus of some biological encoding so
let's say
my i'm triggered into anger but then
that anger takes on the symbol of a
certain
person in a political party or whatever
right there's different
stages of phase change that are
happening there of solidity
that are being offered to that
glass-like structure of order that are
being set up
by those externalities and it seems like
one of the things we just did a
paper on misinformation with a group and
one of the things that's so interesting
to me is it's not
a single event misinformation it's not a

binary

where oh i got misinformation how did i
how could i have prevented it like you
know getting hit on the head with a
stick

there's so many levels at which you can
address misinformation and intercede and
intervene

and this feels like one of those
exercises that perhaps
we can start to identify interventions
for

information integrity by looking at
where

might we put in seed crystals to put in
phase changes at stages
of that process to redirect it
in ways that are more socially or
productive or whatever the goal is of
the group

thank you thank you scott

steven and then anyone else

yeah i think this discussion also it
points to

an important consideration it brings in
the territory and the environment
because a lot of these phase changes
in the world out there but also maybe
unlike say the enamel of our teeth which
is maybe like glass

in a sort of what you would call an
equilibrium

state okay it's at a low energy state

it's just there right i don't expect it
to keep moving that much

but we're talking about these
non-equilibrium steady states

so you've got this kind of micro states

which things can revisit
from which some are more probable from
which
there's attractor states which becomes
these macro states which
then the organism at different
scales is visiting so this is
interesting because there's both going
on
you know i think sometimes everything's
non-equilibrium steady state but
you know i'm holding the computer here
it's pretty thermodynamic equilibrium
object you know i'm i'm in that world
even if my body
has a lot of non-equilibrium dynamics
going on so
that maybe plays out in some of these
questions
thanks steven yep a lot of uh
parallels analogies cross mapping
between
information and thermodynamics
and it's unsurprising i guess because
there's a lot of groundwork there but it
really is a fruitful
intersection maybe we can just remind
ourselves of some of the questions that
we had
from last week and then we can
just continue walking along so anyone
just raise a hand
so uh these were like loosely structured
by
topic but we had big questions how much
mathematics
is required to understand or work with
free energy principle

and active inference and just how shall
we speak
and work clearly on this issue
then we had a few questions on a
philosophical side we recalled
the papers of mel andrews and
anes hippolyto and thomas vanesse which
were
a footnote in this paper as sort of
relevant contemporary work so definitely
on the philosophical side i'm sure we
could
um hear any thoughts about
what next steps are on that line of
research
and this question of how to think about
blankets for groups
you know if it's groups all the way down
then what's wrong with
throwing blankets over groups but what
kind of blankets
and what kind of groups also blue if you
want to
introduce and give any thoughts that's
cool too and
of course questions about what's real or
not
um we had a few themes arising
about again about math and about
how there is um
there's multi-directional dialogue with
active inference and other
fields so what happens when we
cross over between different areas
scott and then anyone else
so i want to talk about markov flags for
a second
and what i mean is the flag of amsterdam

with the three x's
my understanding is from a tour guide
that the x's stand for fire plague
and flood and so
if that let's assume that for a minute
and the idea was if you see that flag
you better follow the club rules on
preventing fire plague and flood
because that's what we do here in this
area in amsterdam
okay so to me that is a markov blanket
because what you're doing is you're
saying hey
we are projecting this out come join the
club
we are doing beijing analysis on how to
prevent cholera
and floods and fire in the city limits
and that's a the signal is there on top
of every building that is has an owner
who conforms to the rules
and so i went i wondered if we can do a
markov
blanket analysis on the amsterdam flag
as a markov flag
expressing and saying all ye who come
here right it's like diagon he's looking
for an honest man i'm looking for plague
preventing and flood preventing people
here so to me that's a
projection out in in in any other kind
of active inference we're projecting out
through a flag instead of sticking our
hand out to
search for a tree so that's active
and then i'm doing regulatory
enforcement
which seems to me to be a body that is

responding to non-conformity
among its members so could you comment
on whether that's a possibility
in terms of projection out of social
elements of markov blankets possible
any amsterdam resident
is welcome to respond uh shannon and
then anyone else who raised their hand
um but yeah so now we're talking about
something like
human of cumulative culture and our
evolution of signs and symbols and how
we
co-construct meaning in space so maybe
the flag if you don't know what it means
it's just the flag
um you might not even know it's a flag
if you see it painted somewhere it's
just something that looks cool and has
some shapes
but if you have been
raised or taken a tour i guess um
in the culture surrounded by the symbol
surrounded by the meaning
all of those small instances of
negotiating meaning and
like we talked about active semantic
inference i can't exactly remember what
that conversation was last week but this
seems like it could be similar sort of
adding meaning and if the flag doesn't
mean to the person you're interacting
with what you thought it would mean to
them you take
action in your world to make it mean the
thing to them that it means to you
um it's an interesting thread i like
that idea

also one thought on the uh flag before
we go to an amsterdam resident is like a
blanket you think about under the
blanket or inside the blanket like
there's a person or a blanket goes on a
bed

but a flag is also fabric and it's
flying in the wind
and the borders it delineates are not on
one side of the air versus the other
side of the air

but as the flag is animated by the niche
through winds

otherwise it's just on the flagpole it
delineates a social

barrier or social um demarcation

which is then cultured in a
intersubjectively negotiated

experience just like you brought up
shannon so yellow and then stephen
all right thanks um it's fascinating uh

i'm not sure i i i know this story yet
from the three crosses i i think i have
a different origin for it but maybe this
is the tour guide so

i'm not quite sure um i

i think this race is also points to to
one of the questions uh

joe was writing in the beginning right
which is in turn like

suppose we can give this analysis of the
flag of amsterdam at the markle blanket
and we can give an analysis of uh the
origin of life in terms of market
blankets

and and lots of things in between right
so what what what does this
multi-applicability

of uh micro blankets say it could one it
could point to a very
deep self similarity that runs through
nature at all possible skills
uh or or could mean that the toolbox is
is in a sense to cause grains to to
so that it applies to everything but but
but you can't really make
uh the relevant differences right i mean
in a sense you would like to think
through what are the differences
indeed through a kind of uh uh
signs of group memberships in in in
social situations versus
uh more biological science versus a kind
of self-organization self-organization
in physics and i think that that's
that's kind of the tension between the
kind of
applicability to everything uh and and
how particularly you would like your
your theory to be
thanks so steven dan scott than anyone
else
i mean this is also brings that question
i think like we just mentioned there
around
where we put things in a markov blanket
where things are part of this
uh external states um hidden states
internal states because in some ways
from the markov blanket perspective the
sensory
states of the epithelia that the kind of
sensory organs is what's been stimulated
and it's inferring an external state in
the world
i.e the flag and that's

having an additional inference that the
flag has got this meaning
another level so um i'm not even sure
what the whole thing's called in some
ways because
is the blanket because the blanket is
dependent on connecting with external
and internal
but it's normally just those that part
of the
of the whole active inference dynamic
so i think that that becomes interesting
is where
what type of uh sensory
input information is within the blanket
biologically
and maybe that isn't quite the same as
what is
thought of when we start talking about
it and that might be good or maybe a
problem
thanks so scott then chris than anyone
else
this is uh this is fantastic this
discussion so
um i realized all of a sudden i thought
oh
wait a second i just realized that some
of these folks may
actually be empiricists and may believe
there is a testable reality
out there i didn't because i'm not that
person
so i i made a living rhetoric
and a persuasive speech right as a
lawyer right
so for me it's all projection so i
realized

i was i was too diminishing of the
question
of whether there exists something out
there to test
and i was looking at and this whole
thing to me this has been a candy store
since a few months ago where i learned
about active inference because to me
it's all um about avoiding
avoidable harms among humans
and to the extent that active inference
is
universally uh assertable
then it has a rhetorical power that's
huge
right so if we can get people thinking
they're surrounded by active
inference blankets and then i say you
know
you don't have an active inference
blanket yet in the way you're doing
business with other people in your
supply chain or you don't have active
inference blanket yet in the way that
you're interacting among nation states
or you don't have active inference yet
in your food insecurity supply chain or
your cyber security all these other
things that i'm working on right now
contact tracing it's information systems
where because they're not within the
same
model then they have conflict
and those most of those are avoidable
harms because they're just humans
working out stuff
that we were separated for a hundred
thousand years when we migrated out of

africa and now we're getting back
together and trying to compare notes on
what we came up with
so the question or this assertion
is that it's interesting i to me
it's not that troubling a question of
whether these
exist or not and i'm and i've read lots
of philosophy and
i've been down the road there but
ultimately
it's the the i say yes and
on on that question of whether it exists
or not to me it's a provisional truth
that's functional sufficiently
functional so i'm not
no longer curious about whether it has
an existence separate from the human
mind
so that so take my comments
with a salt lick not a grain of salt but
a giant block of salt
because i've realized i'm an old old
person and i just don't care
if it actually exists separately and i'm
not saying that
to say this conversation doesn't matter
it's awesome but it's
that that the dynamics of rhetoric
are such that we can employ the thing to
do good
whether or not it's real
so scott has litigated realism versus
instrumentalism
he's serving as the judge jury
executioner lawyer
um chris and then joe then blue
yeah thanks a lot so um i think there's

a

like there's a there's a useful way to
to think of the flag metaphor
so to speak and to bring it a little bit
closer maybe to to also what
scott was saying now and and the paper
itself

so the question that the paper is trying
to answer right

it's kind of to what to what extent
is the pristine blanket construct or the
markov blanket

construct as used by fristen a
conventional

thing so you know there are different
kind of

like you you can have different views of
course about the metaphysics of
mathematics and so on and so forth but
generally there's some agreed rules
about how maths works

or certain branches of mathematics work
and that's where the mark of blanket
construct comes from right it comes from
statistics and

there's a clearly defined way of using
it so the convention for using the mark
of blankets

is very clear it was introduced very
clearly by judea pearl in

his article in later his book at the
break of 80s and 90s and there's no
kind of mystery or you know there's no
question about how it should be applied
productively so

whether or not you you think that
science is after
capturing the reality as it is or

whether you think
science only gives us approximate truths
or some kind of
you know approximate uh
models of one of of what's out there
that that doesn't matter
what's really important for us i think
and maybe the guys can also
speak to that is that for mark of
blankets there's a clear convention
for how to deploy them and it's very
clear
what they are doing and what they can't
give you whereas for
the way fristen applies the notion first
of all he kind of
softens the convention the convention is
less codified
we can think we can say it's less clear
to what extent they apply where and
similarly
it's less clear what you can get out of
it right so this is the
um this is kind of the problem so so
stephen for example mentioned something
about you know boundaries and the fact
that whenever there's an interaction
these ma mark of bl sorry these blankets
let's leave the mark of part of the name
out of it
these blankets appear and or they can be
found there
and it's not entirely clear whether
according to pristine there will be
found
at every uh in every place where we have
an interaction between two systems for
example or something like that right

because the the the the rules uh
in which they are applied are not as
clear as as
in the case of the original markov
blanket construct
and i just wanted to also to follow that
up on this very quickly
in the version of the paper that's under
review now
we actually realized that perhaps in the
draft you've read we were a little bit
uh too harsh on tristan because he does
impose certain rules about for example
how to individuate different kinds of
nodes
but there's still a question of kind of
what
what it means to individuate these nodes
in a certain way
thank you thank you joe thank you joe if
you wanted then blue and scott
i'm fine for now actually thanks thank
you blue and then scott
so without specifying markov blanket or
pearl blanket or fristen blanket i want
to touch back
on what yellow was saying about group
memberships
and also about uh what shannon was
saying about
cumulative culture so i think this is
like a really interesting notion of um
like a markov blanket or a blanket
overlying a group
and and the affordances and like what it
means to be part of that group like age
specific affordances or socioeconomic
specific affordances

under a given blanket and how and when
this like cultural or group membership
prior
gets like instilled and also like do
blankets then overlap
like i'm of this race and of this age
and of this
you know so so these overlapping
blankets i just think it's a very
interesting
um idea about group membership and
cumulative culture and we might know
that we're part of this group
but we don't know all of the specific
underlying
backstory about why we're treated this
way or
were afforded these certain things i
just think that's a very interesting
way to think about it
thank you blue scott and then anyone
else and also anyone who's watching
lives welcome to ask us a question
so it's so interesting couple of things
blue on what you just said so i do a lot
of sewing
and with the notion of quilting is
really
interesting in what you just said
because they're actually a bunch of
quilting museums
that do ethnographies of quilts
and so it really comes back around you
had this quilt which reflects a person's
history
like when they did the hiv quilt each
square represented a person
right so you actually have a physical

membership instantiated in this blanket
and then the blanket being the symbol of
the community being knit together i mean
all these

knits probably not the thing you want to
do in a quilt but anyway okay the the
thing i was gonna

mention in response to joe i think was
joe's comment before about the
oh no sorry it was um it was kristoff's
about the um application

and the alteration from the paper that
notion they have this

um this something that's running as a
scientific a statistical concept that is
brought out in the world

one of the things who as i mentioned
before i work in an applied physics lab
now applied physics is really funny
because you're

applying physics so that seems like
maybe it has something to say about
what's going on here and i

and i realized well what you do in
applied physics and what we do in our
program

in specifically in our program we take
technically feasible systems

so let's say that we could describe
something as technically feasible

the chris the fristen notion is
technically feasible and you can apply
it

in scientific context but what i do in
my

program and what the lab does generally
is then we take the stuff and make it
ready for adoption in the field

so in specifically in my program i take
technically feasible systems and i test
them for whether they're bolt reasonable
and bolts is business
operating legal technical and social
so if something is technically feasible
like the orion project where they were
going to use nuclear bombs to loft
weighty objects into space that's
totally technically feasible
but it would be unreasonable socially
because you'd pollute the entire
atmosphere with radioactive isotopes
right so it works in the lab it's
theoretically good terrific
so what i'm my notion is maybe what's
going on here
is the applied physics
applied statistics and when you get it
out there in the field it's
it can be seen we can call them
compromises or we can call them
realities
or we can call them affordances or we
can call them beijing
adjustments of scientific empirical
views
right which when it gets out there in
the reality maybe the folk understanding
is the reality
of physics in the mind
of the of humans maybe if the mind
exists in society and the brain's just
an antenna tuned into it
anyway just a couple thoughts to play
with thanks scott
so stephen and then anyone else who has
a question yeah i'd be interested

christoph just further in what you said
about
what does it mean to individuate a node
in a particular way in this idea of
nodes and
i suppose at this larger scale um
i'm seen with these partially observable
markov decision processes being used in
some of the modelling they don't
necessarily from what i understand talk
about
nodes in that way but at the primordial
soup
kind of cellular level they seem to
often talk about
nodes in that maybe in in the similar
way or not but i'd just be curious what
you think
thoughts are about what it means to have
a node what kind of node that might be
and what's similar or different between
different ways that this is implemented
yep go for it chris and i have figure 2
up or let me know if you want to go to a
different figure
uh sure thanks a lot i mean those
two the last two comments and the
questions even
really match nicely together i think
because
so you know i mean you can we can talk
about variables right
i mean in a in a graphical model
variables are represented as nodes
basically
so what but what i of course meant is
the differentiation of
the differentiation between active

states and the
passive of or perceptual states inter
inner states
uh yeah exactly well you can also use
the loop but yeah
internal states external states um
these things right so it seems to me
that
uh and i think i mean not only to me it
seems
to all of us that there's a certain
degree of arbitrariness
with regards to what is what i mean at
least in the draft you've read
it seemed like that to us we then
learned
that after really reading
for a lot of tristan's papers very
carefully we actually
kind of extracted the implicit rules
for for how to how to you know label
these variables as active or passive
in a graphical model but
this is still a very arbitrary
distinction one which is uh
like the rules by which you individuate
so i
i used individuation here maybe i
shouldn't have used the word individuate
the way you identify
the role played by a given variable
that's what i should have said
previously is that the rules that
that give you this identification are
still
very arbitrary and they are not actually
anchored
in anything very substantial and this

goes back
to what scott said about the feasibility
and applicability
because of course you can take this kind
of way of partitioning a system
you can build a robot with you know some
kind of
ai system which will be built on the
principles
of the free energy principle and
pristin's
way of thinking about graphical models
and it will be perfectly
you know good at doing certain things
that you build it for
the problem is that when you're building
a system for a particular purpose as an
engineer you kind of hold all the cards
you make all the decisions you know you
decide like how to what like what kind
of active states will the robot need or
something like that
and that's very that's a very different
perspective so to speak from this
perspective
of you know let's call it the what then
it calls the
evolutionary r d uh department right so
an
evolution doesn't really plan things
evolution just throws things that the
wall and whatever sticks kind of stays
and goes on to reproduce and all the
other things kind of fall back
and um in this sense it's it's very it
becomes
very problematic when we think of like
pristine's

grand project of explaining you know
life as we know it for example
as in the paper from which we took this
diagram that's on the screen now
because it's it slowly becomes a little
bit suspicious
whether we can actually whether the
the the way he labels these nodes and
the the kind of the rules for labeling
the nodes
will be the same on for example
different levels of organization
stuff like that right so so whether this
will be applicable
uh to many different biological systems
in the way that he hopes it will be
or not thanks and just to
to clarify um you all wrote that
spectral graph theory is used to
identify the eight most densely coupled
nodes which are defined
as the internal states so for those who
think that given this soup of
interacting particles which are all the
same kind they're all following the same
simple rules
yeah eight densely coupled ones okay
those are internal and then
there's some downstream consequences
with which nodes influence which nodes
but it's hard to argue that choosing the
eight densely coupled nodes is
simply you know true or internal when
you can look at what it is and ask
whether there's a different way to
partition it
so joe and then scott and then anyone
else

yeah thanks thanks daniel that's a very clear way of putting that point um so just to pick up again on something chris chris mentioned briefly there is a further individuation problem i think which is about how to identify which which nodes you have in the first place or which which variables you have and this is a kind of growing question so so you can always you know look inside a node and split up in some other nodes or some other subparts and this this might not be a problem just that active inference suffers from it's going to be any any any kind of modeling uh system is going to have this kind of issue but it is going to make a difference to where the boundaries are drawn and what what kinds of what kinds of things you get so that's just to say that there's there's already a kind of human uh involvement in the model and just choosing what scale you pitch your nodes at thanks joe uh scott and then dean yella and stephen so in that notion love it this is great and that clarification about the nodes being assigned in a sense contextually in a given system state um it seems like if we start to that may be a pathway to revealing the self other ch or starting to unpack some self

other challenges so maybe
a node that is once characterizable
as internal then characterizable as
external
is revealing something about the paradox
of
membership in groups that
the in fact it's not the node that
changes
but the sensation of whether itself or
other
so you start to take that always
paradoxical this is that hofstetter
stuff and everyone else who talks about
when does a group become a
when does a group form when does it have
an internal voice when does it speak an
extra voice all that stuff
and so maybe that analysis and that
reassignment possibility starts to get
into that
and maybe that also informs blue's
notion before about the group
you know kind of the again that i'm
calling it now the markov quilt
as of that last conversation but that
group identity
blanket where it's patched together
maybe those things are
like irving goffman and things like that
or mirror neurons where you want
something to be recruited by others
as an internal state you project it to
them as an external state
intrinsically and so the communication
you're inviting
others to encode their markov blankets
similar to yours in a sense or something

like that
so that is exciting to me that
slipperiness
of assignment there feels like something
that offers a pathway to analyzing
those dynamics of group and individual
identity thanks thank you scott so
dean yellow and stephen and blue
yeah um so back to construal
and the question of how do i get the
world i think
a lot of the conversation today has
centered on maybe a more clunkier
expression which is
how do us get the world either
us as flag conformists
or us as red dots versus cyan dogs
um so even pristin
stephen shared a video where he was
talking to the santa fe institute
even pristin said that things get a lot
more interesting when we get to the
combinatorial level
and they're not the same as when we're
talking about
how do i get the world that there is
there is a difference when complexity
arises so
can we hold up pearl and mark off
blankets and
get past the ore of this or that
because we recognize that it's a little
bit different when
how do us get the world that's a
construal question
and so that's what i'm kind of curious
about
thanks dean really powerful and a big

difference between how do
us get the world how do we get the world
how do i get the world
how do they get the world these are like
really markedly different
questions so yellow stephen blue
um so i was thinking getting back to
blue's question
and shannon's questions on group
membership but maybe if somebody else
wants to
uh well i i i i i just get talking on
a group membership um
so i mean to what extent one the kind of
marker blanket formalism applies
to to groups uh i think it's an
interesting one
and i think that's one thing that really
matters here is is really how you
conceive of a group right i mean you can
lead like a group of people i mean we
are a group of people that are spread
out
all over the world and you could think
like a group in a more narrow sense like
a flock of birds
or a
these are entities in the same kind of
spatial temporal
location that coordinate their their
behavior with one another and you can
think of
as a predator arriving also as a
perturbation to that system as a whole
right so in the way that entity is
coordinating it's
it's it's it's dynamics with with the
rest of the environment you might

actually think that no this is a group
that actually too weak
the active inference principles do apply
and you might in that sense might think
as well that the group as a whole
somehow has a
as a as a as a marker blanket
um i think it's that another analysis is
more difficult to give if you think
about
uh group memberships
uh also kind of the graded group
memberships right i mean
uh i think blue was talking about uh
yeah we belong to multiple groups at the
same time
uh and not actively acting on being part
of that group but just as a kind of
label that i would
uh endorse uh when you ask me i think
there the application of markup blankets
is a lot more difficult and
at most more metaphorical so i just want
to say
so exactly how to understand groups
there is is is
is is is important if if shannon wants
to understand
kind of crowded damage or so that that
might be more applicable
and if you want to really understand
kind of social membership and graded
membership
and membership of different groups at
the same time then uh
yeah i find it more difficult to
understand how kind of the active
influence framework can be applied to

those kind of groups thanks
so steven blue scott
yeah i think so sticking with that group
idea that
there could be the the idea of
morphology and the amount of constraints
and affordances in the the whole
environment body organism dynamic that's
constraining
and where maybe there's not that
constraint so i'm thinking i know
someone that she's trying to do some
markov blanket meditation processes i'm
kind of thinking if someone's meditating
and
starting to just sense their states in a
maybe a
a way that slowly is like a cloud that
self-organizes
in a way i could see a more plausible
way where
sort of something like not
obviously anything the same as this kind
of markov
blanket in a soup could be emergent
however when you're in a you know a more
real cognitive environment where you've
got these priors
and and the only way we're able to
access it anyways through self-reported
data which we a much higher level of
dimensionality
it may be is is only in some of the
temporal processes that we can get some
analogies
like but i think that yeah it may be
very different at different scales
so the way that constraints play out i'm

wondering how we think about constraints
um and not having constraints
nice question steven are the constraints
hooked into this
phrasing and then or where are the
constraints and how does that
shape how we model things across
different scales so blue
and then scott and then anyone else
so hopefully it won't go on too long but
i want to touch back on what stephen
just said and also what um
yela said you know when i think about
how
group membership goes into active
inference like how can we relate these
two things i think about
like specifically what is the likelihood
that i'm gonna take a ride on a yacht
today
right like so my socioeconomic status
doesn't really
afford me the opportunity to ride around
in yachts very frequently so i'd be very
surprised
and i also live in the desert so i'd be
very surprised
if i was riding around on a yacht at
three o'clock this afternoon like that
would that just doesn't go into my
model so perhaps our group membership um
goes into you know construction of
of the prior um and our model in ways
that are
maybe layered right like you know an
onion maybe
um and so then i want to touch back on
what scott said because this is what

really made me raise my hand about group alignment
and we were starting to talk with casper hesp when he was on the stream about
um inferring the states of others so
i think this group alignment i wonder
about this a lot and and
uh specifically with respect to
communication which we've talked about
in relation to active inference and
cooperation
and you know building a collective
intelligence so when we think about
you know whether the earth is flat or
not right like this at one time was like
a hotly
debated topic and maybe still is but um
this is like there's been a convergence
right so we all now align
our like most of the people in the world
agree that the earth is round
so i wonder about this time to
convergence and this has been used to
measure
information transmission and can we
similarly move it through
estimate collective intelligence so like
when we there's a time to convergence
for
a swarm or this might also represent
like a phase transition
right so there's a time to convergence
and if it happens rapidly like the idea
is readily adopted that now the earth is
round
so i wonder if these divergent views
that we have
as people in groups in divergent groups

in a society
that do they hinder our collective
intelligence
or do they like actually assist it
because without having divergence
we could never actually come to the one
true answer
so is it resistance that hinders the
evolution of intelligence or
is it um i just it's something to think
about
thanks blue um scott if you're
go ahead yep i just repeat the question
you asked right there at the end
yeah repeatedly yeah so i just wonder
about
our um the divergent views
right that we hold in society about
whether the earth is flat or round
do does this divergence help or hinder
our collective intelligence and the
evolution of intelligence overall
or is it resisting adoption of a new
view
right like so perhaps there's there's
resistance to change
even when we see something is correct
like climate change we can clearly see
the effects of climate change
but there's a huge resistance to
changing
so is it this resistance that hinders
our collective intelligence like what
where is the evolution of our collective
intelligence helped or hindered
by divergence and views or resistance to
adopting new views
sorry goes along yeah shane you want to

give a thought on that and then scott
yeah um so if we were going to
talk this would be like whether a system
is
resisting it's a non-equilibrium steady
state and like
moving to a different steady state um
and
i guess that's a question i don't i like
i've been seeing pop up in a lot of the
papers recently is how can you define
not every system is going towards like
a minimized free energy gradient or not
every system is
trying to maintain a stable state like
systems dissipate and
re-emerge and dissipate and re-emerge um
so i don't know is that just a way to
re-ask that question
but talk about something a bit more
physical
than than collective intelligence
um and i guess collective intelligence
maybe
is like collective agreement on a
certain
set of of ideas or norms or something
rather than like
intelligence like smarter than another
person
is what we're maybe talking about here
so how does that system arrive on this
steady state of what we think that the
world is
thanks shannon and there's the
convergence on shared
generative model on shared mindset but
then it made me think of

ants colony relocation and over
evolutionary time
colonies are shaped to appropriately
relocate
but not instantly and not just choose
the first um
you know first aid coming back with
exciting nest location you can't go for
the first one
so how to actually bear with
the process of distributed computation
or distributed intelligence
knowing that it looks like disagreements
at multiple levels especially when it's
not just as simple as ants
relocating but cultural values and
beliefs and everything like that
can i jump in again about aunt yeah yeah
oh yes
there was a new paper and i don't know
who was by i saw it on twitter
um but it was about ants they travel
across a scaffold and it's either it's
like
flat and so they just walk or they tilt
it and so some ants start to fall down
and they start to build this scaffold
um and then when they build a scaffold
ants can all walk over it so they're
supporting each other
um and i'm thinking um
i might be diverging from our markov
blanket topic here but i'm thinking
um that collective intelligence is like
as long as the world
looks flat as long as the world is easy
for us you know to navigate
then we can all go about our day we

don't really need to try to converge
with anybody
but as soon as the world starts to be
disrupted slightly so we turn that
scaffold
folks start to fall off like then we
need to build
something build a shared collective
notion of reality and then
then the convergence matters and so that
could be something like as soon as
there's a perturbation to the system
then it's useful to try to
recognize boundaries or try to
have some sort of intervention i think
scott was mentioned to
have a boundary that's suitable and
helpful and productive
and that could be where it's fine to
diverge all you want as long as
everything's simple and then as soon as
it gets complicated
or we start to like have harm for people
then we need to
have a system that can
result in some sort of convergence happy
for someone to bring me
back on topic it's it's a great point
and i almost interpreted as how do we
help those at the fringe who might be
being harmed
how do we make that the stimuli and
um for realignment and for
re-scaffolding rather than keep on
cutting off more and more oh the failing
parts of the bridge
they gotta go they gotta go well now we
have no one on the bridge

instead of seeing that those on the
fringe and can be part of
the recovery so scott kristoff and
steven
so shannon that was right in the middle
of the discussion there's no
you don't have to bring back anything
that was perfect so
admiral rogers the former head of the
nsa once said don't let any
emergency or disaster go unexploited and
the idea is
that the disaster is that situation
people act in a disaster you know
hurricane
sandy whatever people you know starbucks
they were putting out power strips for
people to share whatever right so
it's exactly that and the rhetoric the
projected
projection rhetoric can cause a state
change
right the anticipation of a disaster
when people are actually experiencing
they go oh my god
this might happen to me or neighborhood
watch right people kind of you know they
go in and they want to protect
property and things so i love that that
point
and bringing it with the other points
before i think it was kristoff that was
asking maybe somebody else about
you know that active inference in a
group
and does it dissipate in the activity if
you're just a member of a club and it's
kind of

lying dormant is that really describable
with active inference
and it occurs to me you know if you look
at group dynamics it may still be
usefully described that way and here's
here's how i would think of it
if i uh this is irving goffman's stuff
that i
i project myself out as i want to be
perceived by others
so if i join a club i may join a club
because i want to be
like i used to be part of an entomology
listserv i've never been a
this is not about enterman's cakes but
insects you know
but i was part of a lister back in the
old listserv days and i just was
fascinating
i sort of wanted to be involved in it
right so i was projecting myself out and
i started making these connections with
folks
so in a way you can de-risk and leverage
by being a member of a group even
passively
right i can i'm extending out my feelers
out there into the world by having that
group membership
i'm de-risking by having those that
group available to me as me being a
member
if it's a group that you know so so i
wonder
if the dissipation of activity
itself is an appropriate gauge of
whether you're a member of a group or
not

maybe in certain leverages and certainty
riskings

there's a functional engagement that's
maybe more passive by the individual but
still may stand for something in the
overall

goals of that entity that's that's one
point and the other one other point
is the the leveraging the risk in groups
i keep coming back to eukaryotes
and eukaryotes you know we it's a better
deal we got multi-cellular got a lot
more options

we got mitochondria we got chloroplasts
we got all the good apparatus instead of
having just these

compartments where you kind of do stuff
in a single cell organism right

so it's a better deal and so maybe part
of this thing with

the groups this goes back to blue's
notion

is is it are we resisting i think it was
a blue

maybe but are we resistant to is this
resistance

or facilitating the group membership
what's going on here

and you know the divergences and the
differences and i started thinking about
thermodynamics first law if you don't
have if you have isothermal you can't
perform

cannot perform work if everyone has
equal information

markets don't have trading

so there may be a benefit to not having
to having divergence right and maybe

in fact um the
that reality is there isn't a
facilitation or resistance it just is
and that paradox is what we're is the
reality we're seeing is that it doesn't
facilitate anything there's no goal
orientation individuals may have goals
but that's a
more bounded goal than the super system
of biology
anyway just a couple of thoughts that i
gleaned from that last bit and also just
to remark once again what a fantastic
conversation
thanks scott so kristoff then stephen
and then anyone else
um yeah so i just wanted this
well what did i want to say uh i guess
two things really
so one thing goes back to blue's comment
about
like the fact that the fact that if you
have some kind of a
complex network there will be multiple
markov blankets in there so it's
completely
feasible to understand you know
membership to certain groups or
something like that or some kind of
relationship structures
to be overlapping in different ways so i
think
it's conducive to this but i think that
we should not
like we're very focused now on this idea
of
you know blankets as um
as a way of intervi like of thinking

about some kind of boundaries again
between groups perhaps or between
different kind of units of organization
and we're not like it's it's it's a it's
always useful i think to kind of go back
you know zoom out a little and not think
about the blankets because you have the
blankets whenever you have some kind of
a network
of you know variables um
interconnected in some way but what's
really interesting is that
in many cases the
behavior you will get from these systems
if you
project them onto the social domain will
be very surprising
so one really surprising finding is
something called the solman effect
it's with a z and double l and it's
basically
a finding that shows that if you have
bayesian
agents interacting in a
network if the network is fully
connected they won't
actually come to a consensus
about some true belief so so imagine you
have a network of
fully connected network of bayesian
scientists and they're
testing two hypotheses right and they're
testing these hypotheses they're
increasing their credences and they're
communicating their credences with each
test
to their neighbors and what's really
surprising is that

the talmud effects shows us that these scientists will not actually converge to uh to a singular credence in in the true outcome

what is also surprising is that a less connected network will actually eventually converge to the to the to kind of to all the members having a very high credence in the you know most likely outcome so this kind of i just wanted to kind of throw it out there

because i think that it's you know we're thinking now intuitively about these blankets as membership to groups and stuff like that but the kind of interactions we might get from using this kind of formalism and projecting it to a social domain might be quite surprising

right i'm not saying it will be useless because perhaps the talmud effect to go also back to what has been said about the flat earth situation

and you know the belief in the roundness or flatness of the earth the solemn effect actually tells us that in a hyper-connected society like the one we are living now there it's even rational agents might fail to converge to the to the you know the most likely hypothesis right and that actually is something interesting

that it's in a way that solomon effect
predicts the
appearance of certain conspiracy
theories in a hyper-connected society
because

uh given the demographics for example of
flat earth

uh belief it has been declining for
decades

until the advent of internet so it only
came to a sharp rise with the with with
social net with social media when people
started to communicate in a much more
interconnected way so i'm not saying
that

you know thinking about these systems in
a social domain is bad

no i just wanted to bring this

let's say anecdotal effect to

to say that uh the the kind of

behaviors we will see these systems uh

converging towards might be very

surprising right or we we might

they might not work exactly the way we

think

they do on like lower scales of

organization

thanks for those points so steven and

then

anyone else yeah that's some good points

there actually and

i think this is where active inference

um

and the mark the fristen blanket has

some advantages

over the pearl blanket if it's about

what action it can

there's a risk you see the information

you start to have this thing that's all
information driven rather than action
policy driven
because it and by keeping it with the
action policy
um for instance one of the things about
having
us on the planet in the solar system
which we used to have
well now we know in a galaxy in a milky
way blah blah blah is that
people have this sense of look we almost
meant to look at ourselves from space
now
which is really only a proviso of modern
technological
perspective people might want to be able
to stay grounded
in some way there may be actually a
psychological challenge
with having to take this outer space
perspective of us on a little ball going
around the sun
you know so there could also be an
action policy perspective here around
flat earth gives you a way to re-ground
yourself because it's like well i'm on a
flat earth so
i don't have to take that i don't have
to go and imagine myself in space
looking at myself
so there's a lot of interesting aspects
to
the action policy side once you start to
have this scaling
and that that could be interesting um
to to to keep that in mind
thank you steven joe and anyone else

uh yeah thanks um just something you
said for steven uh when you say there's
an advantage to the first and blankets
over the pearl blankets
do you mean an advantage you know for
real systems making use of these things
or an advantage for us as modelers
this is something that i really worry
about sometimes that we slip into
thinking about the practical benefits
for a system
of of these constructs whereas what
we're really concerned with is
the benefit
it's for accurate and and in that
context both might be good tools but
when you use one tool or the other will
depend on what you're trying to do with
them
i would say more for how we look at it
as a modeler
although the actual thing gets called a
system and then has this kind of
perspectival
process i think it's good to keep it in
mind
so that action doesn't get lost and it
doesn't purely start to become
information yep great now that that
makes sense and um
i wasn't necessarily accusing you of
this error but sometimes i do worry that
people uh
people get mixed up over who who's
getting the advantage i guess you know
who's
who's benefiting here but but if if you
mean just that it's important to keep

to keep action as part of the picture
and part of our models that that's a
fair point

yeah and one contribution of this
paper and we look forward to the um
revised and published version as well is
like

it's almost like if all we know is
markov blanket

it's like hey hand me the screwdriver
well there's different shapes there's
different sizes if you have
the hex it's not going to work for the
phillips and different countries or
different applications

need different screwdrivers so now we
can

with these adjectives ahead of blanket
we can start to differentiate hand me
the phillips screwdriver okay
you're talking about the fristen blanket
you're talking about the pearl blanket
and so

we can be specific now because it's
almost like enough air was pumped into
the

markov blanket metaphor it was becoming
too big of a tent

too big of a blanket and so now we can
start to be really specific and and
address

those kinds of questions when are we
talking about utility for the system
when are we talking about utility for
the modeler which of course
maps very closely to um the
distinguishing

in uh that you raised with inference

with an inference
within a model so scott oh yeah
if any of joe if you want to give a
thought on that and then yeah
can i can i just i'm really so we're
already pleased with the reception this
paper's had this particularly my
reactive inference community and just
just the reason you you said that i
think
one thing we were worried about but
people have seen is that there is meant
to be this more positive message that
these aren't necessarily bad tools
they're just different tools for
different kinds of
cases so i'm i'm really glad that people
have seen that and kind of taken that as
a
helpful message it's it's the yes
and approach to the literature it's not
blood sports it's not a knock down drag
out with citations
we can just build upon each other and uh
come to more
useful and good ways of communicating
so scott and anyone else yeah just two
points one along those lines
in in law one of the most effective
strategies for disambiguating something
is saying what it isn't
right and so even if you don't know what
it is you can at least start to say what
it isn't
and so that's a handy notion along the
lines here and as
last couple of comments we're observing
we had this thing out there that was

very exciting
and we kind of everyone was playing with
it and said oh yeah and it's the same
things happened in privacy
the word privacy is just carrying water
for all sorts of anxieties right now
the word trust has all sorts of
precision so
the problem with language is we got a
lot of subtlety and a lot of
interactions interactions are
increasing exponentially through
exponential increase in interactions
globally is an assertion that i make at
least
and if you think about that we still
have a vocabulary that's
the same as it was way down the
exponential curve of interaction
increase
so how do we capture our subtleties in
our interactions
right how do we expand our vocabulary
what's exactly what we're talking about
here is we take words
we use them by analogy in other areas
right we've already talked about
blankets well these aren't really
blankets at all and in fact we get
hijacked by the notion of blanket we've
already seen that
it's starting to use those is it a
blanket or is it a membrane
well is it semi-permeable well a
blanket's not really semi-permeable you
know that kind of stuff right
so it's fascinating that we
have this is the the appropriate and

um very exciting challenge to have
to the taking it and run running with it
idea
early on in disambiguation and it means
that there's a lot of energy which means
there's a lot of need
for whatever it is that's being served
by the concept
and so what you're as modelers this gets
kind of interesting kind of gets into a
manhattan project kind of ethics of
science right
because as modelers you do have
responsibility and you're discharging it
here by saying hey
what does this mean for society you know
should we develop nuclear energy
unfettered with regard to whether it can
be used as a weapon
right because this is this is powerful
stuff here and so that
raising of the notion and the
questioning of whether it's appropriate
to apply these things
socially is going to come up very
quickly as
people do things like the rhetoric that
i'm talking about and start to apply it
politically
and start to recruit biases using these
concepts
that will be seen as a nefarious
application of a scientific and
mathematical statistical concept
and so your paper and the questions
you're raising now
are starting to lay that foundation of
pause and saying to people in the future

hey

don't be so fast to run with this and

just take a minute

to think about the implications what

isn't this

right and so starting to ask that

question can be as important and

sometimes more important than what it is

thanks

thank you there's a question from the

chat

sean kelly wrote are there special cases

of markov blankets that shield

internal states a markov shield so

would any of the authors like to just

give a response to that

yeah yellow and then go ahead stephen i

think

i can give it a try um

of course this really uh depends on what

you mean by shielding

uh right and uh i think in the paper and

also

i mean in the paper we use a statistical

notion of shielding

that is pretty much equivalent to

conditional independence

right so if

if you have three nodes and there's one

intermediate node

such that knowing the intermediate node

will

will mean that knowing the third

node doesn't

not mean anything doesn't give you

anything

for for knowing your first node uh then

that's a notion of shielding in this

in in the sense that the second node
kind of epistemically
shields off uh the the
activity of the at the third node from
from whatever happens at the first node
and this is an epistemic notion of
epistemic and statistical notion of
shielding and also that one right i mean
shielding has a lot of kind of more
everyday kind of connotations that are
necessary not necessarily can be
congruent with this physical
uh notion but if you mean a statistical
notion then
yes the very definition of a markov
blanket is
that the things inside of the blanket
are given the marker blink it's shielded
off from the rest of the world in that
statistical sense
thank you that was what i was thinking
insulation like a
heat insulator is shielding it and so
that is kind of what these
nodes are doing statistically and then
we're nuancing what kinds
of of categories these nodes might fit
into
and that's a key difference between
pearls phrasing and fristen's phrasing
is the extent to which nodes are
classified into different kinds
so steven and then anyone else
yeah i think that question about how
things and when we scale up
is really important and in predictive
processing and the modeling you know you
have this hierarchical model

of things going up to higher levels and
yeah there could be this implicit
suggestion that each of those steps or
somewhere up there it's going to other
size

blankets but it may not be that blankets
are needed

as much um to go another layer it might
be that there's

some sort of process structure that
is doing the job at lower levels it's a
more

active influence process and if i could
give one metaphor there was a metaphor
i'd put on about

if there's a flag stuck in some mud and
it looks quite stable but it's blowing
in the wind

okay and then i have another flag which
is stuck in some mud it's actually the
top of an

ant hill and underneath the ants are all
swarming to hold it still

so and it's actually a little bit more
still but actually the hole is more open
so in the first case it's energetically
thermodynamically more an equilibrium
is being held in a sp in one place

the second one you think it's in more of
a thermodynamic equilibrium because it
seems more firm but actually there's a
whole swarm of ants underneath
trying to hold it in place and but none
of them have got that intention but
looking from the outside the system is
holding the flag

but from the point of view from the ants
they're trying to enact some sort of

action policy and i suppose i wonder if there's something similar between you know the system as seen from the outside as a state and the kind of swarming that's going on underneath and that the two can get mixed up sometimes

thanks steven um always a great uh challenge an eternal question what are the ants trying to do but what you just pointed out there is is really it's related to this distinction they draw with these four

different cases the difference between a generative model which is like the modeler's perspective on a something versus the generative process and so

those are two different generative processes that are giving rise to the movement of these two different flags let's say there's two different processes underlying it

those are realist claims about what is giving rise

and then there's all different kinds of perspectives that can be taken

um but if we mix up two people's perspectives or if we mix up a generative model a generative process it's a category error and if we're coincidentally

correct it would be miraculous because it'd be a

the wrong kind of thing that's being considered

so it is extremely important to um

prevent that kind of misinterpretation

by really

asking are we talking about the process

degenerative process or the generative

model

and sometimes even just that

distinguishing as well as this four-fold

uh

classification raised here these go a

long way towards helping

model so scott and then anyone else

since we have europeans on the line i'd

like to talk about kant for a second and

i want to

display my ignorance of kant

um and the categorical imperative

so i i i complain about the categorical

imperative

because i if can one person

know enough have enough situational

awareness so that their good acts

can translate into goodness in the world

and again maybe that's total

misinterpretation of con

or maybe just got it all wrong but in

the same way

i wonder about markov blankets and

this notion of knowledge of an awareness

so that because because that one of the

problems we see in the world is even a

good actor

if they don't have good in information

can do bad things like in the

sustainable development goals of the

united nations

they put in a gujarat power plant was

funded by a un

based financing agency and it cost a

billion dollars or whatever
and they were crowing about the success
of sustaining
sustainable development goals of energy
and development were
being pursued and furthered but what
they didn't
talk about and they then got sued in in
the
court over is the fact that the local
fishermen and local population was
pollution
and so it poisoned the local waters and
the local population with the pollution
so they didn't do a sufficient analysis
even in this situation where they're
trying to improve the world
and listing out all these goals of the
world that's their
declaration that's their amsterdam flag
of what they're trying to do with the
u.n
right so but nonetheless they pursued
one goal
at the expense of other goals and one of
the things i'm
interested in your thoughts you can go
into contact one and also
correct me on my bad county and
non-interpretation there
but um can you comment a little bit on
awareness and knowledge
we talk about group and individual
markov blankets but can we talk a little
bit about
that notion about sharing of knowledge
and situational awareness
just as a survival strategy the ants

were surviving in the
um shannon's tilting bridge
example before um which was
a survival thing in stephen's flag
projection that was a little bit more
the answer a little bit
smarter about their externalities than
that one
anyway but can you comment a little bit
about how do we get
how how might these analyses and these
models
and their implementations in reality and
in the lab
help us to understand the nature
of group information float thanks
thanks for the question so if anyone has
a
thought yep gala go for it and then
anyone else
great i i'll give it a try i'm not sure
if i can tie it all the way back to the
categorical
imperative but that's uh let's give it a
go
um so i think the first thing
that the way i understand it uh the kind
of active inference framework is really
a theory of learning for a particular
kind of
environments so in a sense organisms
start to learn to adapt to the
specific niche niche that they are that
they are
engaged in and then somehow
slowly build up a particular model of
that particular kind of niche
that you can of course and that that

model you can use also
for hypothetical kind of situations or
counterfactuals or
or new kind of situations but there's no
guarantee that this kind of previous
history
uh uh i mean if you if you if you enter
a new niche or if the world changes
it's it's not guaranteed that your
previous knowledge
will be uh uh will be specific to the
new situation and we'll be we'll work in
that
in a new particular kind of situation um
so i guess there that indeed the the
kind of
active inference framework gives you a
perspective
on
situated situation specific knowledge
and not on the kind of categorical the
domain general knowledge and so this
doesn't
guarantee indeed that if you use your or
use your model to make inferences about
new situations that that you will
end up on on the right result and so
that gap between what will be your best
action
according to whatever kind of imperative
you set for yourself in reality and how
it works out that's always a gap in in
in in the knowledge that you have
uh what you need is related to the
difference between the kind of the
causal dynamic of the world
the generative process and the general
model of the world that you

that you have and so i get that that

that's indeed

why perhaps i mean speculating that that

different kind of

ethical frameworks might might might fit

better in such a kind of approach rather

than

[Music]

the other logical one

that was very oh yeah go ahead chris

chris

uh yeah i just wanted to follow up on

this a little

bit because we the the notion you know

the the distinction between the

generative model and the generative

process was

brought up before and now yella alluded

to it and i think it's really important

to keep in mind that

the free energy principle so so the

generals maybe let's start differently

the generative process right

is supposedly the process or the the the

causal process that produces certain

sensations in an organism

and then the generative model is

supposed to be

on this framework producing predictions

right which which kind of explain away

in the bayesian sense

uh the differentiations that the model

is the

the the organism is experiencing

and the idea that's implicit in a lot of

the literature is that you know you have

the generative process on this side you

have the generative model that's trying

to mirror the generative process there's
the sensory sheath in the middle
and that these two these two things will
somehow you know fall down to each other
and they will be just perfect
uh there will be some kind of a perfect
resemblance or mapping between them
um and that's not entirely it's not
entirely clear that
this is the most successful evolutionary
successful strategy so
the generative model only needs to
recapitulate the generative process to
the extent that it's successful
for in a given ecological niche or even
social niche
to ensure the survival and reproduction
of the organism right
um and so so that's one thing that's i
think important to
remember about this that that you know
the free energy principle
from the get-go kind of does not
necessarily
produce agents that are perfectly
that will be always um optimally
informed about what's going on
in their surroundings they just kind of
really they they learn just enough to
stay alive
and of course the inclusion of active
influence gives them an advantage over
traditional bayesian agents because they
can also go and try things out
and learn in this way but it still
doesn't
give take us all the way so to speak uh
to to to kind of transcend

our model and and and and get to know
the the process that
that that's out there in the world and i
think this is also true in the social
domain in many ways so
so in the social domain i think it's
very plausible that the same thing
happens
and we could expect that the same thing
will happen for i
know groups for example uh
where where groups will form certain
models about how
society operates or how the world is
which will be optimal to their
given epistemic situation and they're
given social situation but will not be
objective uh models of how the how
society or the world operates
thanks and what you just said there at
the end about
the cultural ways of viewing the world
it's um
known that many cultures their own name
for their own
country or homeland might be like this
is the middle spot or this is like the
main place
it makes sense and it it's um
it's unsurprising that it's that way but
then
just like scott said we're all comparing
notes that's the phase of our
test that we're in and two people
are mapping to a different territory
with the same
relative egocentric map and so this
question about how different egocentric

perspectives

and at which level uh those egocentric

perspectives exist at

how those come into harmony is really an

interesting

question so stephen and then anyone else

who'd like to speak

and yeah this is awesome times it's been

a great discussion we have

20 minutes or however much more people

would like

yeah i'd just like to i really thought

that was a really interesting point

there about

when other approaches are maybe better

than taking an ontological approach and

i suppose it

i think and you're talking about the

epistemological approach

i suppose that you get into this

question of whether it's

a more fuzzy i'd be interested in your

thoughts on like fuzzy logic and a more

fuzzy based epistemology um

rather than maybe the type of logical

categorization of knowledge which ends

up becoming ontological at some point

and that there is this idea of

ontological pluralism

but they're sort of their way around

this you know or

third wave is that the third wave

inactive consciousness

has that this idea that the mark of

blankets because it breathes with

different regimes of attention you

it's sort of trying to get around that

problem but

there is it doesn't necessarily say how
they do that

so just be curious yeah a bit more and
and the other thing if it's tying in

with other areas is like for instance

they're talking about affect now a lot

and affect is this kind of integration

which then ties in in qualitative

research where you have axiology which
is kind of the the

normally this thing which is sort of

down there in arts-based methods but you
know the

the one where it's about how people feel

about things well now that starts to

become

part of a lot of what we're

understanding about the world so it

could actually have quite an implication

for participatory interactive kind of

research methods and making them

much more central in a way so i wonder

any of your thoughts on that

about about that from the authors

thanks steven if anyone has

any thoughts on that or can raise their
hand

yeah yell it go for it um

i mean so so i guess what what we do in
our paper is a little bit

before that that step right so uh on the
question of

uh uh ontology phases versus

epistemology

should you go for a kind of ontological

pluralism

or not i think what you tried to say is

that

luke here's a is a class of models that
are typically used in literature
as abstractions from a target phenomenon
right
and so if you want to enter if you want
to understand those
uh uh in kind of on the logical terms
that that's
that's fine but you need to give a story
about how
how that abstraction relates to the
target phenomenon
uh uh that is that that is there so
either
uh if you go to into metaphase you go
from a kind of more
uh platenese kind of root and you say
well actually this mathematical model is
not an abstraction from the target
system but it actually
nature at its jones is somehow
mathematically structured
um or you you say look no these are
these are indeed abstractions
produced by us and they they're useful
in predicting and interpreting the world
uh but they are not themselves the the
the world
uh so in a sense we we kind of offer the
the kind of dilemma to anyone who
who wants to make a commitment or a
choice there between epistemology or
ontology or anthropological pluralism
um saying like look you you can go the
ontological route
and uh of course this is the way more
challenging interesting kind of exciting
uh route but then you either need to

bite this

this this bullets that that these models

are not really

abstractions but somehow underlying

reality or need to give some some

other sort of story about how that

how it works and uh i mean i'm i think

personally i'm curious to see what what

what what the response from from the

community will be

uh which of those uh options people will

take but

i mean i don't think either of us is

necessarily committed to one of those

those which are not but just saying okay

look if you want to do it then these are

these are the rules kind of so to say

maybe disagree with the rules that's

fine as well but then give an argument

for

that

thanks very interesting stuff

anyone else have any thoughts

uh yeah stephen and then anyone else go

for it

yeah i think that i i agree a lot with

what you're saying there i think there's

there are some gaps

in um and and rather than saying there's

a gap sometimes

it's just filled with the markov blanket

if that makes sense

um if in doubt fill it with a markov

blanket

and i've been really curious i'm

actually looking a lot

in terms of how organisms structure

like the idea of parry personal space

and and spatial
inactive ecological approaches and and
the way that
maybe there's a a way that that space
is actually um and there's a field
called mental space psychology which has
been a very fringe
field um actually a lot of it based in
holland and that
um is uh it seems more plausible now
with this so it was a much more practice
based approach but
um i thought i'd share that anyway
because i'm presenting something this
friday so
um but it's i i do think there's
something missing in terms of
um mechanisms that extend out
beyond the body into the space and it
may not have to be
um at all levels pristine's
blankets in in the traditional sense
shannon and then anyone else
um i'm gonna talk about
brains so i know we've been
extending you know into the world into
the body into the body and everything
um and we've been talking a lot about
group
dynamics and maybe how groups behave or
how we feel like
part of a different group um and
earlier we super briefly touched on
mere neurons um so these are neurons
that fire when
i observe an action the same as when i
do an action
and there's been a lot of work lately

not necessarily on near neurons
but on inter brain synergies or you
might call them interbrain
coherence or cross brain coherence where
lots of dynamics in my brain are firing
very similar to lots of dynamics in your
brain
um and this could be because they can
work from mary hassan
on neuro cinematics this could be
because we're watching
a film and it's
evoking similar responses because we're
all attending to the same stimulus
but the film that we're watching
it could have a structure where the
filmmaker sort of
designs what you pay attention to um and
so
most people's gazes will fall in that
direction um
you'll have even more similar sort of
inter-brain coherence
or unstructured where you're just sort
of filming a scene and there's maybe
some people over here some people over
there and
your gazes will could differ a little
bit more these structures could differ a
little
more um and
this is all to say how
like the film is the environment in this
case
and
if we're talking about the rest of the
world being the environment or like us
in this room being the environment our

gaze is structured in a certain way by
there's a blue box on my jitsi screen
so it's sort of directing everyone's
attention at my face
um but there's also some
lack of structure i don't know that
we're seeing the same
tiles in every space
and there's different ways that
if we interacted like if i interacted
with one of you
a lot in my daily life like maybe steve
and i hang out and get coffee every day
and we talk about something every single
day so we have
a propensity to pay attention to similar
things in similar ways
and we've been specifically talking
about yella's idea and how much we agree
or disagree with it or something
so we are continually like drawing our
gaze to yellow to see if he understands
um or agrees with what we're saying like
there might be this similar inter-brain
coherence between stephen and i
that's different between me and blue
because
blue and i never talk we're not even
friends um these are all hypothetical by
the way
um and these so a question it's not you
know um
in your paper necessarily but like if
we're looking
for markup blankets in the world
one thing that we could look at is
something like a
inter brain coherence and that might be

something real in the world or it might
be a pearl blanket something that we use
to describe how people
interpret something the same or how
their brain activities
are synergistic in a certain way
and if you can be
empirical i know we were being
rhetorical a lot of times but if you can
be empirical
and look at these signatures of either
you know conversation dynamics or brain
dynamics
and can relate them to how the world is
structuring them
then that's a way that you can look for
you can still keep looking for
markup blankets in the world or find it
useful
um it's definitely i think
what we keep coming back to when we're
talking about
um these blankets as a metaphor these
blankets is something real is that we
can totally have
both as long as we're specific about
which one we're referring to
thank you shannon yeller and then anyone
else
thanks shannon that's really interesting
um
and i think that there there there's
there's a couple of things there right
i mean and and in either one sense if
two people
have the same same background same
experiences all right they might they
might

parse a particular kind of movie or
scene in in
in in a similar kind of way such that if
you put them in in room separate from
each other
you you will indeed see a similar kind
of neurodynamic
going on in both both cases maybe tight
indeed to attention where you pay
attention your gaze etcetera
um the other one is indeed kind of
on-site kind of interpersonal
synchronization that might be happening
right and i know there's this kind of
it's kind of second person neuroscience
kind of approaches that that try to
capture those kind of things in the
experimental setting
and see how how how in what particular
kind of settings
brainwaves synchronize and uh et cetera
um and that kind of second perspective
reminds me a lot
of uh uh of the kind of discussion
that's there in active inference on
generalized synchronization right so
these are the kind of examples that you
get when as there's uh
uh two coupled clocks standing on the
table and is tend to synchronize after
after a while so this is a very simple
causal mechanism
uh causal structure where
synchronization just occurs over
or over over time um
and when i first started thinking about
uh uh marco blankets
i i i used that kind of example as a

kind of reduction that
absurd him saying like look if if these
things even have a marker blanket right
then it's an empty
uh uh empty notion now
having worked with these with these
philosophers of science here i think i
think
need to be a bit more more more specific
and indeed say
look okay whether that thing has or has
not a michael blanket that that that's
that that's a different question but
there's there is a point that there is a
very
uh uh simple structure of kind of
causal dependence and causal
independence such as if you would fix
the table
perfectly the things would not
synchronize and if you would let it
slide a little bit and wobble a little
bit then they would they do synchronize
so there is a kind of thing
like this intermediate factor that
mediates a causal relationship between
those
uh uh those two things now the question
is
is that do you need to have a markup
blanket for that does mark
do market blankets add anything there or
is it kind of
a story about kind of causal dependence
and causal independence
enough i think that well
causal dependence there and cults are
independent such that if i would do this

then there would be no synchronization i
think that's enough and then
what do you call that a kind of markov
blanket
is is a second kind of question you need
to translate it back to the kind of
world of
causal models and and see how they
relate to them uh
but i think there's yeah there's a clear
case to be made that this is parallel
between
the kind of cases you are describing and
the kind of synchronization settings
that are there
in in this kind of couple clock examples
thanks and one interesting aspect to
kind of
bring it back to this discussion about
the internet you know what kind of
beliefs propagate culturally on what
kind of connectivity uh
just like shannon was saying with the
video chat and with the live stream
what somebody sees it's not just like
they're sitting in a different chair
at the theater and they're watching the
same play or they're in separate movies
uh you know rooms watching the same
movie and so cultural cues might lead
somebody
to look into a correlated position as
another person
but um when there's the customization
it's like each person is in their own
theater
which takes it to another level because
it goes even

it's actually different inputs and the
inputs can be targeted in
interesting or new ways so
it's kind of stretching the limits of
our social cognition
because it's not just a bunch of people
asking what they just saw with a shared
event
it's like trying to come to a shared
understanding when people are seeing
different events
so yeah there's some new
fun things we're learning on the
internet i guess
well in the last couple of minutes
if anyone has some thoughts i'd be
curious
to the authors or non-authors where do
we
go from 20.2 where's our
momentum or direction or curiosity
one thing i'll bring up while someone's
raising their hand is
uh without uh buzzing
in the middle of a conversation how can
we be really clear
when we're intentionally or
unintentionally slipping the difference
between a markov for pearl frist and
blanket
and generative model generative process
with a model within a model
we don't want to police subtleties
of language but at the same time
somebody can really
build a bridge that leads someone to
nowhere
if they're unintentionally uh not sure

the difference between these concepts so
i'm just curious
how do we promote uh rigorous and
accessible norms around talking about
these ideas
recognizing that we're all like totally
learning this and
and formalizing the project around us
yela and then anyone else
uh yeah just a small uh uh note and
i'm not i don't remember exactly who who
mentioned this but
uh at some point we talked about kind of
two phases
of research right and one is kind of
taking a lot an id
and running along pushing it along as as
far as you can
and then there's a new kind of phase
where like okay now we ended up in a
completely new
new uh how how how are we gonna kind of
substantiate this how can i make kind of
formalizes a little bit
and i think this is the the kind of the
i guess over the last few years
active inference knows the philosophy of
active influence has kind of been just
and also i mean
also in my own earlier papers kind of
been running as far as possible see how
far we could get this and see what would
happen
as a kind of creative kind of kind of
startup stage of of research
and i think both in this paper and then
a number of other papers that are
currently

kind of come coming out um
uh i think this kind of more kind of
consolidatory fake consolidatory phase
is is happening where we're like okay
but okay this is where we ended up does
it really make sense
need do we need to make some more
fine-grained demarcations to
to do we need to tweak the conceptual
apparatus to to
to have it all make sense uh and that
it's a different kind of work but but i
think that that's that's that's really
important in the kind of
for the next kind of phase for for for
active inference uh research so i hope
these kind of discussions can contribute
to uh uh to doing that kind of work
we're consolidating bitcoin to the key
20 000
level you know let it rest before the
next uh
maybe whatever metaphor works for people
but um any other
thoughts because that's very nice yellow
thanks everyone for sharing this was an
awesome sequence
and it really helped take our our lab
and our
discussion to another level we would
always welcome any of you to come back
to continue the conversation come for a
guest stream
come as a visitor to um be on any
conversation you'd like
so thanks again to everyone and for
listening
peace